

CTU-Chat: Supervisor-Routed Multi-Agent RAG with Heterogeneous Knowledge Allocation for University Counseling

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Abstract. University counseling spans relational curricula, cohort-specific fees, exemption policies, and narrative regulations, each requiring distinct evidence representations. Whether decomposing this heterogeneous task into specialized agents improves reliability over an equivalent single agent remains underexplored. We present CTU-Chat, a supervisor-routed multi-agent system assigning queries to four bounded specialists with scoped tools and representation-specific evidence paths: Neo4j graph traversal for curricula, structured lookup with deterministic calculation for financial data, and hybrid retrieval with neural reranking for regulations. We evaluate CTU-Chat across three scenarios on a 100-query held-out benchmark: (1) architecture-level comparison against a matched single-agent baseline and a routed generic variant under identical capabilities; (2) architectural ablations removing specialist policies, tool isolation, heterogeneous evidence, deterministic calculation, and route repair; and (3) a semantic-neighbor tool-ambiguity stress test across 11, 31, and 51 tools. In architecture comparison, CTU-Chat achieves a 10.0 pp higher Automated Fact Pass Rate ($\text{Pass}_{\geq 50\%}$) point estimate (76.0% vs. 66.0%; paired 95% CI [+2.0, +19.0] pp; exact McNemar $p=0.0414$) with 41.4% fewer recorded specialist input tokens, alongside an unlogged upstream routing overhead of ~650 tokens and sequential dispatch latency (10.65 s pipeline vs. 5.16 s post-retrieval baseline). Under registry expansion to 51 tools, the scoped specialist sustains 88.0% tool E2E Success (vs. 78.7% for Top-10 selection). These findings indicate that domain-scoped specialization and heterogeneous evidence allocation enhance institutional counseling reliability, albeit with measurable trade-offs in coordination latency and token consumption.

Keywords: Multi-Agent Systems · Supervisor Routing · Architectural Ablation · Heterogeneous Knowledge Allocation · Retrieval-Augmented Generation · University Counseling

1 Introduction

At the opening of the 2025–2026 academic year, Can Tho University (CTU) reported 48,960 undergraduate students¹. Its scale and evolving regulations create advisory needs spanning heterogeneous knowledge forms: (1) **academic curricula** encoding hierarchies and prerequisites; (2) **tuition schedules** varying by cohort and field; (3) **exemption and scholarship policies** combining eligibility conditions with rates; and (4) **administrative regulations** comprising lengthy narrative documents. Prior systems address individual domains—CAAS for IT admission counseling [9] and REBot for regulation advising [8]—but a single representation or retrieval method may be unsuitable across all domains.

RAG and GraphRAG augment LLMs with documents and knowledge graphs [7,4,2], while agentic architectures support tool use and multi-agent coordination [19,17,15]. These approaches do not prescribe how heterogeneous institutional evidence should be represented, routed, and processed. Known limitations in language-model arithmetic [13] further motivate deterministic calculation for financial queries.

Central Hypothesis: When a counseling system must handle diverse knowledge forms and semantically adjacent tools, routing queries to bounded specialists with representation-specific evidence paths may improve reliability relative to a single agent with equivalent capabilities, albeit with coordination latency. Whether this trade-off is favorable requires direct comparative evaluation.

These challenges highlight three primary research gaps: (1) limited evidence comparing multi-agent and single-agent architectures under matched capabilities; (2) limited mechanistic understanding of which components drive observed differences; and (3) limited evidence on how bounded specialization maintains reliability as the tool registry grows. We formulate three research questions:

- **RQ1:** How does the proposed multi-agent architecture compare with a functionally matched single-agent in end-to-end reliability, evidence grounding, and cost?
- **RQ2:** What performance changes are observed when specialist policies, tool isolation, representation-specific evidence access, deterministic computation, and route repair are individually removed or replaced?
- **RQ3:** How does bounded specialist routing affect tool-selection reliability as semantically overlapping tools expand the registry from 11 to 51 tools?

This work provides three contributions: **C1**—a supervisor-routed bounded multi-agent architecture with four specialists, deterministic route repair, and partitioned tool privileges; **C2**—an architecture-level comparative evaluation against a functionally matched single-agent baseline and a routed generic-worker variant; and **C3**—architectural ablations of five components and a semantic-neighbor tool-registry stress test across 11–51 tools.

¹ CTU opening ceremony report, 5 September 2025: <https://en.ctu.edu.vn/news-t/can-tho-university-inaugurates-the-2025-2026-academic-year-embarking-on-a-new-era-and-affirming-its-role-as-the-leading-institution-in-education-training-science-and-technology-in-the-mekong-delta.html> (accessed 28 September 2026).

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the CTU-Chat architecture. Section 4 describes the experimental setup. Section 5 reports results. Section 6 discusses findings and limitations. Section 7 concludes.

2 Related Work

2.1 Vietnamese NLP and Retrieval Foundations

Vietnamese is an isolating, tonal language where white spaces separate syllables rather than words [10]; 85% of word types contain at least two syllables. While domain-adapted bi-encoders capture semantic paraphrasing, Vietnamese higher-education decrees rely on rigid Sino-Vietnamese legal terminology and alphanumeric cohort codes, which necessitates the coupling of dense embeddings with exact lexical matching.

BM25 [14] provides reliable exact token matching for rigid identifiers but cannot capture paraphrases. Dense retrieval [6,7] captures paraphrastic intent but may struggle with rare alphanumeric tokens. Reciprocal Rank Fusion (RRF) [1] merges BM25 and dense candidates without score calibration, while neural cross-encoders [11,18] jointly encode query–passage pairs for reranking. Adaptive-RAG [5] demonstrates complexity-based retrieval selection. These approaches motivate dynamic retrieval but do not allocate institution-specific relational, tabular, and narrative knowledge to isolated specialist paths.

2.2 Knowledge Graphs and GraphRAG

Knowledge graphs represent domain knowledge as structured triples [12]. GraphRAG [2] leverages graph topology to uncover multi-hop associative paths that escape flat chunking, benefiting curriculum structures and prerequisite chains. However, universal graph extraction may introduce spurious relationships, motivating selective graph construction restricted to domains with well-defined relational schemas.

2.3 Agentic RAG and Multi-Agent Orchestration

ReAct [19] demonstrates interleaved reasoning and action, AutoGen [17] provides a multi-agent conversation framework, and Toolformer [15] studies tool invocation. For institutional counseling, these capabilities raise design risks without explicit constraints: broad tool exposure increases errors; peer-to-peer exchanges increase coordination cost; a shared index mixes evidence across domains. Controlled comparisons against functionally matched single-agent baselines remain limited, as most evaluations report absolute performance without isolating the contribution of multi-agent decomposition from underlying capabilities.

2.4 Educational Counseling Systems

Educational chatbots assist in advising and learning support [16]. CAAS supports admission counseling for IT programs [9], while REBot combines retrieval with category-guided GraphRAG for academic regulations [8]. CTU-CHAT departs from prior systems across three aspects: (1) *Task scope*: modeling 111 curricula as relational property graphs with deterministic arithmetic for tuition; (2) *Multi-agent routing*: a LangGraph supervisor routes to four bounded specialists with isolated evidence spaces; and (3) *Architectural evaluation*: direct comparison against functionally matched baselines with component-level ablations.

2.5 Research Gaps

An analysis of the reviewed literature reveals three primary gaps: (1) limited architecture-level evidence comparing multi-agent and single-agent systems under matched capabilities; (2) limited mechanistic understanding of which components drive observed differences; and (3) limited robustness evidence under tool ambiguity as semantic neighbors accumulate. These gaps motivate the evaluation design in Sections 4–5.

3 Proposed Model

This section presents CTU-CHAT’s architecture, algorithms, and evidence-acquisition mechanisms. The design follows three principles: *centralized agent coordination* (a contract-checked supervisor dispatches each query to one domain agent); *heterogeneous knowledge allocation* (each information structure maps to a corresponding evidence representation); and *representation-aware evidence routing* (graph-tool or intent-scoped retrieval path according to the assigned domain).

3.1 Overall Architecture

Figure 1 illustrates three coordinated tiers: (1) the Interaction and Centralized Supervisor Tier, (2) the Domain-Specialized Multi-Agent Core, and (3) the Heterogeneous Knowledge Storage and Tool Tier.

CTU-CHAT specializes LangGraph through typed state (`AgentState` with validated rewrite, routes, evidence, and response), an intent-owner contract that checks structured supervisor output before dispatch, conditional routing (academic queries enter the graph agent directly; others pass through retrieval), acyclic orchestration (specialists return to END without peer chatter), and asymmetric tool registries (scoping tools per specialist). In live deployment, CTU-CHAT maintains multi-turn session state via Redis and applies validated query rewriting. For the offline experimental evaluation in Section 4, the system is evaluated in a controlled single-turn benchmark harness with fixed queries, isolating the routing, retrieval, and tool-scoping mechanisms without query-rewriting confounds.

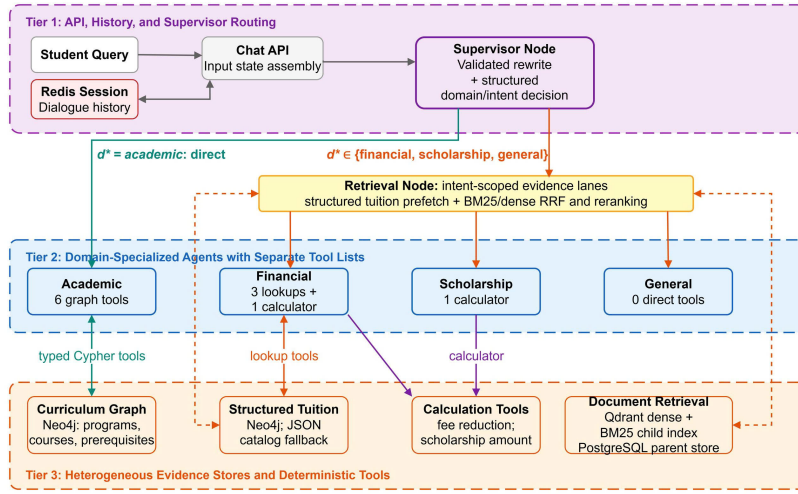


Fig. 1. Implemented CTU-CHAT architecture. Redis history enters the graph state; the supervisor’s structured decision is contract-checked before conditional dispatch. Academic queries enter the graph-enabled agent; other domains pass through intent-scoped retrieval.

3.2 Neo4j Knowledge Graph and Cypher Tools

The Neo4j graph is built from 113 curriculum Markdown files representing 111 distinct program codes. The resulting graph stores approximately 7,000 nodes across 10 labels: curriculum entities (e.g., Program, Course, Block, PLO, PEO, Track, Job) and tuition entities (TuitionFee, TuitionPolicy, ExemptionRate) connected by over 16,000 edges across 13 typed relationships encoding containment, prerequisites, program outcomes, tracks, occupations, tuition, and exemption bases.

The system does not ask the LLM to generate arbitrary Cypher. After the supervisor selects the academic agent, its ReAct policy chooses one of six typed tools and supplies arguments. Each tool normalizes identifiers and calls a fixed, parameterized Cypher query implementing three query classes: *attribute lookup/search* (program codes, names, comparisons), *structural traversal* (prerequisite chains up to ten hops, course intersection, reverse path search), and *structured financial lookup* (tuition, policies, exemption rates). This design binds agent actions to deterministic graph operations, reducing exposure to errors from unconstrained Cypher generation.

3.3 Routing and Deterministic Arithmetic

For a query q and its dialogue predecessor, the supervisor first attempts a validated standalone rewrite, then predicts domain d_{raw} and intent ι_{raw} under a structured schema, falling back to `general/other` on parsing failure. The ownership contract resolves rule conflicts or incompatible domain–intent pairs. Academic queries proceed directly to the graph agent; other domains build intent-scoped retrieval lanes, with financial queries first attempting Neo4j tuition lookup and then JSON fallback.

Deterministic Tuition Arithmetic. For fee-reduction requests, the tool `tin_h_toan_hoc_phi` receives three parameters—per-credit tuition T , per-credit exemption basis B , and exemption percentage $r \in [0, 100]$ —and calculates deductible credit amount $D = B \times (r/100)$ alongside net payable tuition $T_{\text{net}} = \max(0, T - D)$. The tool clamps T_{net} at zero. The primary contribution is the integration of validated rewriting, constrained routing, and deterministic arithmetic into a unified multi-agent workflow.

3.4 Heterogeneous Knowledge Allocation and Hybrid Retrieval

Rather than a uniform document index, CTU-CHAT preserves relational curricula in Neo4j, accesses cohort-dependent tuition through structured lookup, and retains narrative regulations in document retrieval. Academic queries enter the graph agent directly; others enter retrieval. Financial retrieval first attempts Neo4j tuition lookup with JSON fallback; the repaired intent then selects metadata-filtered document lanes.

For narrative regulations and scholarship criteria, BM25 and Qdrant search 400-character child chunks; parent sections (2,800-char target, 100-char overlap) are loaded from PostgreSQL and fused via RRF: $s_{\text{RRF}}(d) = \sum_{s \in \{\text{bm25, dense}\}} (60 + r_s(d))^{-1}$. The full stack pools intent-governed and unfiltered candidates, deduplicates, and reranks with `bge-reranker-v2-m3`. Scores below 0.20 are removed unless fewer than two candidates remain; graph evidence is added under fixed quotas and up to seven passages are packed for generation.

3.5 Cross-Domain Query Handling

To handle queries spanning multiple domains, CTU-CHAT adopts a *single-owner with signal-based evidence prefetch* policy: the supervisor selects one primary owner based on dominant intent, and the retrieval layer expands evidence lanes when keyword signals indicate related domains—for example, adding academic-program or academic-regulation lanes when curriculum terms are detected, a tuition lane when fee-related terms appear, or a scholarship lane when scholarship keywords are present. Specialists do not invoke other specialists; if the query requires evidence outside the owner’s scope, the signal-detected lanes prefetch it through retrieval rather than cross-specialist delegation. This design trades full cross-domain tool composability for bounded execution with targeted evidence expansion. The 20 cross-domain held-out queries directly test this mechanism.

4 Experimental Setup

4.1 Dataset and Annotation

We evaluate CTU-CHAT on 200 administrative queries constructed through a document-grounded protocol over official CTU statutes. The 100-query development split supports index calibration and prompt engineering. The 100-query held-out set follows a predefined distribution: 40 direct, 20 within-domain multi-hop, 20 cross-domain, 10 entity-comparison, 5 temporal, and 5 adversarial queries (51 formal, 49 colloquial; 38 single-source, 62 multi-source). It contains 19 Academic, 28 Financial, 17 Scholarship, and

16 General single-domain queries; the remaining 20 combine evidence across domains. Candidate queries and ground-truth evidence were extracted from official statutes: one annotator established reference answers, required facts, and gold sources, while a second annotator independently verified all annotations; disagreements were adjudicated against official regulatory text. An audit found no cross-split duplicates (max 5-gram Jaccard = 0.1875). Each query is annotated along nine dimensions, including domain, evidence type, and tool requirements, to support fine-grained failure analysis.

4.2 Baselines and Configurations

Baseline equivalence and execution bounds. For Scenario 1, all architecture configurations share the same LLM (Gemini 2.5 Flash Lite, $T=0.0$), knowledge representations, retrieval stack (BM25 + dense + RRF + bge-reranker-v2-m3, $k=7$), and deterministic calculator access. This setup is designed to isolate the effect of the orchestration architecture. Scenario 2 intentionally changes one component at a time. To reduce retrieval variance in Scenario 1, all three architecture variants evaluate over the identical shared full-stack evidence retrieval output, yielding matched pre-generation Source Recall (78.6%) and Source AP (70.8%). The shared evidence context excludes outputs of tools whose invocation is evaluated; tool-derived results are available only after successful tool execution. Differences in final answers may be associated with the combined changes in tool privilege scoping, prompt specialization, and agent routing; Scenario 1 does not isolate their individual effects. Multi-agent variants use intent-routed scoped registries with identical execution bounds: maximum 3 tool calls, matching context limits, 2,048 output-token budget, and 30-second timeouts. Token accounting in the offline harness records specialist agent execution; supervisor routing introduces an estimated upstream overhead of ~ 650 input tokens per query (derived from the supervisor prompt template, structured routing schema, and average query length) in multi-agent variants. Reported input and output token comparisons between S1-C and the single-agent baseline reflect specialist agent context scoping, and are explicitly qualified alongside total pipeline costs.

Scenario 1 — Architecture Comparison (RQ1): Three configurations are evaluated on the held-out set ($N=100$ queries, 300 runs under greedy decoding $T=0.0$): **S1-A:** Unified Single Agent with all 11 tools, no supervisor or routing; **S1-B:** Routed Generic Worker—supervisor routes to domain workers with scoped tools but a single generic prompt; **S1-C:** Full CTU-Chat with supervisor, contract repair, specialist prompts, scoped registries, and deterministic computation.

Scenario 2 — Ablation (RQ2): Five ablations each remove one component from Full CTU-Chat ($N=100$ queries per ablation, $T=0.0$): **S2-A1:** No Specialist Policy (generic worker prompt); **S2-A2:** No Tool Isolation (all 11 tools visible); **S2-A3:** Uniform Text-RAG Baseline (graph data serialized to text, traversal disabled); **S2-A4:** No Deterministic Calculation (LLM generates arithmetic directly); **S2-A5:** No Route Repair (raw supervisor output used directly).

Scenario 3 — Tool Ambiguity (RQ3): Fifty fixed cases are evaluated while 40 semantic-neighbor tools expand registries to 11, 31, and 51 tools. Four configurations are compared: Full-Registry Single Agent, Global Top-10 Single Agent, Routed Generic, and Routed Specialist. Each case uses 3 counterbalanced tool-order blocks, yielding $50 \times 4 \times 3 \times 3 = 1,800$ decisions across counterbalanced blocks.

4.3 Evaluation Metrics

Metrics span four groups. *Final-answer quality*: Required-Fact Coverage (continuous proportion of reference facts matched via substring containment in generated response), Answer Correctness, Answer Relevancy, and Faithfulness (Ragas [3]). *Evidence quality*: Source Recall, Source AP, Context Recall, Context Precision. *Execution reliability*: Tool Invocations per Query, Tool Selection Accuracy, Argument EM/Contract Match, Result Accuracy, and route repair rate. *Operational cost*: input/output tokens, LLM call counts, and wall-clock latency. Source Recall and Source AP serve as controlled retrieval diagnostics, identical by design across Scenario 1 configurations.

Primary Evaluation Metrics. In Scenarios 1 and 2, the primary decision endpoint is the *Automated Fact Pass Rate* ($\text{Pass}_{\geq 50\%}$), defined programmatically as the proportion of responses where at least 50% of predefined required fact phrases are matched via automated substring containment ($\text{FactCov} \geq 0.50$), complemented by continuous Fact Coverage. Since this lexical metric measures surface phrase occurrence rather than semantic or arithmetic accuracy, it functions as a surface-grounding proxy. To address this limitation, it is complemented by independent deterministic evaluations: in Scenario 3 by *Tool E2E Success* (correct tool selection, valid parameters, and error-free execution), and on the arithmetic subset ($N=9$) by numeric exact-match auditing against official bursar schedules.

4.4 Statistical Protocol and Measurement Scope

Evaluations use greedy decoding ($T=0.0$) to minimize sampling variance. Paired differences across configurations are computed at the query level ($N=100$); 95% CIs are estimated using 10,000 bootstrap resamples. The primary confirmatory hypothesis compares CTU-Chat versus Unified Single Agent on $\text{Pass}_{\geq 50\%}$. Timing follows configuration boundaries: baseline S1-A latency measures agent invocation post-retrieval, while multi-agent variants measure pipeline latency inclusive of supervisor routing, route repair, and specialist execution. Reported tokens record specialist agent execution; upstream supervisor routing adds an estimated overhead of ~ 650 input tokens per query (derived from prompt and schema length).

4.5 Infrastructure

Gemini 2.5 Flash Lite via Vertex AI ($T=0.0$). Neural reranking on a GTX 1650 (evaluation hardware). Neo4j 5.20 stores relational curricula; PostgreSQL stores parent chunks (2,800-char, 100 overlap); Qdrant indexes child chunks (400-char, 50 overlap) with a 768-dimensional vietnamese-bi-encoder. System uses bge-reranker-v2-m3, LangGraph StateGraph, and Redis 7. Latency excludes the first warm-up run. Prompt templates, tool contracts, and graph schema are archived.

5 Results

5.1 Scenario 1 — Architecture-Level Comparison

Table 1 compares the three configurations on the 100-query held-out benchmark. Because pre-generation evidence retrieval is held fixed across configurations, Source Recall (78.6%) and Source AP (70.8%) are identical across all three architectures. The observed differences in final answer quality and operational cost may be associated with the combined orchestration changes described in Section 4.

Table 1. Scenario 1: Architecture-level comparison ($N = 100$). Pass₅₀ = Automated Fact Pass Rate ($\geq 50\%$ required facts). Tools/Q = mean tool invocations ($\downarrow$). In/Out Tok = specialist agent tokens. Bold = best reliability / lowest cost.

Arch.	Pass ₅₀	Fact	Src.R	Src.AP	Tools/Q	In Tok	Out Tok	Lat. (ms)
Single Agent	66.0	57.5	78.6	70.8	0.41	6296	1530	5159
Routed Gen.	66.0	54.2	78.6	70.8	0.31	3473	1415	8921
CTU-Chat	76.0	61.0	78.6	70.8	0.28	3690	2092	10652

Pairwise bootstrap analysis (10,000 resamples): (1) CTU-Chat vs. Single Agent: CTU-Chat achieves a 10.0 pp higher Automated Fact Pass Rate (Pass _{$\geq 50\%$}) point estimate (76.0% vs. 66.0%, paired 95% bootstrap CI [+2.0, +19.0] pp; exact McNemar $p=0.0414$), alongside higher Fact Coverage ($\Delta = +3.58$ pp, 95% CI [−1.2, +8.5] pp) and 41.4% fewer specialist input tokens (26.1% fewer total specialist tokens: $\Delta = -2,044$, CI [−2,710, −1,378]; Single: 7,826 vs. CTU: 5,782; supervisor adds ~650 input tokens/query), under differing measurement boundaries (10,652 ms end-to-end multi-agent pipeline vs. 5,159 ms post-retrieval single-agent generation). (2) CTU-Chat vs. Routed Generic: specialist policies raise Fact Coverage by +6.85 pp (CI [+0.7, +13.0] pp, $p < 0.05$). (3) Routed Generic vs. Single Agent: both configurations yield an identical 66.0% pass rate, while Routed Generic exhibits lower Fact Coverage (54.2% vs. 57.5%, $\Delta = -3.27$ pp). This parity highlights two distinct failure modes. The Single Agent suffers from decision space contamination and prompt interference: presenting all 11 tool schemas simultaneously dilutes context attention, introducing cross-domain tool ambiguity and unnecessary invocations (0.41 tools/query in Table 1). Conversely, Routed Generic eliminates decision space contamination by restricting tool visibility via the supervisor, but its unspecialized worker prompt lacks domain-specific reasoning directives and output templates. Consequently, tool scoping without specialized prompts produces shallow answer synthesis, failing to improve factual fidelity (54.2% Fact Coverage) and demonstrating that tool isolation and specialist prompting must operate conjointly.

On the 20 cross-domain queries, CTU-CHAT achieves an 80.0% pass rate (vs. 70.0% Single, 65.0% Generic). Evaluating sensitivity across thresholds, CTU-CHAT demonstrates a higher automated pass rate at the baseline 50% fact-coverage threshold (unadjusted exact McNemar $p=0.0414$). This margin narrows at 60% (54.0% vs. 50.0%) and reverses at stricter thresholds (e.g., 27.0% vs. 30.0% at 75%). However, the difference

at 75% remains statistically uncertain ($p=0.5488$, 95% CI $[-10.0, +3.0]$ pp), indicating neither architecture holds a definitive advantage under stringent fact-matching criteria.

Trade-off Analysis and Operational Boundaries: Across configurations, Table 1 demonstrates distinct architectural trade-offs: (1) *Accuracy vs. Latency*: CTU-CHAT achieves the highest factual reliability (Pass_{.50} = 76.0%, Fact = 61.0%) and the fewest tool invocations (0.28 tools/query). However, this accuracy requires an end-to-end dispatch latency of 10,652 ms. In contrast, the Unified Single Agent optimizes response latency (5,159 ms), operating at over twice the speed by bypassing intermediate routing and contract validation, albeit with lower pass rates (66.0%). Within multi-agent execution, the +1,731 ms delta between Routed Generic and CTU-CHAT corresponds to specialist prompt instructions producing +677 additional output tokens (2,092 vs. 1,415) for domain-tailored reasoning and evidence formatting. In ablation S2-A4, disabling deterministic calculators reduced latency by 4,540 ms. (2) *Token Economy*: Specialist prompt scoping reduces recorded agent input tokens (3,690 vs. 6,296 in Table 1), while supervisor routing introduces an upstream overhead of ~650 input tokens per query in multi-agent variants. (3) *Deployment Context*: These contrasts indicate distinct deployment roles. Monolithic single-agent designs suit low-latency, general institutional FAQ workflows where response immediacy takes precedence. Conversely, CTU-CHAT is preferable for high-stakes counseling involving tuition arithmetic, prerequisite chains, and multi-policy regulations where factual errors carry administrative consequences.

5.2 Scenario 2 — Architectural Ablation

Table 2 removes one mechanism at a time from Full CTU-Chat ($N=100$). Deltas are relative to the full system.

The two largest observed degradations occur in *No Specialist Policy* ($\Delta\text{Pass}_{.50} = -10.0$ pp, 95% CI $[-19.0, -1.0]$ pp; $\Delta\text{Fact} = -6.85$ pp, CI $[-13.0, -0.7]$ pp) and *Uniform Text-RAG Baseline* ($\Delta\text{Pass}_{.50} = -9.0$ pp, CI $[-15.0, -4.0]$ pp; $\Delta\text{Src.R} = -12.3$ pp, CI $[-16.5, -8.3]$ pp). Removing tool isolation causes no change in point estimate ($\Delta\text{Pass}_{.50} = 0.0$ pp, CI $[-6.0, +6.0]$ pp) while expanding input tokens by +1,270. Disabling route repair yields $\Delta\text{Pass}_{.50} = -2.0$ pp (CI $[-7.0, +3.0]$ pp), disproportionately harming cross-domain queries (-15.0 pp, dropping from 80.0% to 65.0%). Disabling deterministic calculation produces $\Delta\text{Pass}_{.50} = -4.0$ pp (CI $[-9.0, 0.0]$ pp) across the benchmark; on the arithmetic tuition subset ($N=9$), exact-match answers drop from 8/9 to 3/9 (-55.6 pp exact match pass) due to unconstrained arithmetic hallucination. Collectively, the ablations distinguish two functional roles. First, specialist policies, heterogeneous evidence, and deterministic calculators drive accuracy at the 11-tool scale. Second, tool isolation (S2-A2) acts primarily as an efficiency mechanism (+1,270 input tokens, $\Delta\text{Pass}_{.50} = 0.0$ pp); its contribution to accuracy is further examined at larger registry scales in Scenario 3 (Section 5.3). Query family breakdown in Panel B reveals distinct performance profiles across domain modalities: while cross-domain queries reach 80.0% pass rate, graph-based curriculum queries record 63.2%. This gap reflects the structural exactness demanded by curriculum queries, which require multi-property credit aggregations across degree blocks (general, core, elective, and prerequisite chains), where omitting a single credit constraint leads to automated lexical failure. In contrast, cross-domain queries

Table 2. Scenario 2: Architectural ablation results ($N = 100$). Panel A reports overall benchmark differences (Δ relative to Full CTU-CHAT). Panel B reports Automated Fact Pass Rate ($\text{Pass}_{\geq 50\%}$, %) across query families.

Panel A: Overall Benchmark Ablations					
Ablated Component	$\Delta\text{Pass}_{.50}$	ΔFact	$\Delta\text{Src.R}$	ΔTok	ΔLat
No Spec. Policy	-10.0	-6.85	0.0	-217	-1731
Full Tool Vis.	0.0	+0.93	0.0	+1270	+296
Uniform Text-RAG	-9.0	-2.18	-12.3	-300	-1910
No Route Rep.	-2.0	-1.27	0.0	-219	-1610
No Determ. Calc. [†]	-4.0	-0.93	0.0	-782	-4540

Panel B: Fact-Grounded Pass Rate ($\text{Pass}_{\geq 50\%}$) by Query Family				
Configuration	Graph ($n=19$)	Struct-fin ($n=28$)	Narr-reg ($n=33$)	Cross ($n=20$)
Full CTU-Chat	63.2	75.0	81.8	80.0
No Spec. Policy	52.6	67.9	72.7	65.0
Full Tool Vis.	68.4	67.9	84.8	80.0
Uniform Text-RAG	63.2	60.7	78.8	60.0
No Route Rep.	68.4	78.6	78.8	65.0

[†]Full benchmark with calculators disabled. $\Delta\text{Tok}/\text{Lat}$ in tokens/ms per query.

benefit directly from the route-repair mechanism, which resolves domain dispatch conflicts and maintains evidence coverage across disparate policy documents (evidenced by the 15.0 pp drop from 80.0% to 65.0% when route repair is disabled).

5.3 Scenario 3 — Tool Ambiguity Stress Test

Table 3 reports the semantic-neighbor experiment across 11–51 tools ($N=50$, 3 counter-balanced tool-order blocks, 1,800 primary decisions).

As the registry scales from 11 to 51 tools, the Full-Registry Single Agent’s E2E success rate drops from 86.0% to 83.3%, while its input tokens increase by 124%. The Global Top-10 changes from 83.3% to 78.7% E2E, alongside a Gold Recall change from 100.0% to 96.0%. The Routed Specialist records 88.0% E2E at 51 tools with 9.8 visible tools. At the 51-tool endpoint, Routed Specialist has a +9.3 pp E2E difference over Top-10 Single Agent (95% CI [+2.0, +17.3] pp) and a +4.7 pp point-estimate difference over Full-Registry Single Agent (95% CI [-1.3, +11.3] pp) with 52.4% fewer input tokens (1,770 vs. 3,722). The relative difference in performance degradation (Specialist vs. Top-10) is +7.3 pp (95% CI [0.0, +15.3] pp).

Analysis of stage-level timing across the 1,800 decisions reveals that Routed Specialist maintains a consistent latency profile across registry expansions (2,091 ms at 11 tools vs. 2,116 ms at 51 tools). Here, supervisor routing introduces a fixed overhead of 946.5–961.8 ms (~45% of total decision time), tool gating requires 1,051.6–1,079.3 ms, and tool execution takes 71.0–92.8 ms. In contrast, while the Full-Registry Single Agent avoids supervisor dispatch (0 ms routing overhead), its gating latency increases from 1,098.1 ms to 1,272.2 ms as prompt schema tokens scale from 1,663 (11 tools) to 3,722

Table 3. Scenario 3: Semantic-neighbor tool ambiguity ($N = 50$, 3 counterbalanced tool-order blocks).

Reg.	Config.	Vis.	Gold-R	Sel.	E2E	Tok	Lat
11	Full-reg single	11	–	94.0	86.0	1663	1098
	Top-10 single	10	100.0	93.3	83.3	1541	1148
	Routed gen.	5.1	98.0	90.7	80.0	1204	1889
	Routed spec.	5.1	98.0	93.3	85.3	1441	1983
31	Full-reg single	31	–	91.3	84.7	2706	1157
	Top-10 single	10	98.0	90.0	81.3	1149	1033
	Routed gen.	9.8	98.0	88.0	79.3	1396	1958
	Routed spec.	9.8	98.0	93.3	88.7	1830	2015
51	Full-reg single	51	–	92.7	83.3	3722	1289
	Top-10 single	10	96.0	86.7	78.7	1036	997
	Routed gen.	9.8	98.0	91.3	82.7	1340	1907
	Routed spec.	9.8	98.0	92.7	88.0	1770	2021

(51 tools), whereas the Routed Specialist bounds prompt schema tokens between 1,441 and 1,770 across the same expansion.

5.4 Supporting Retrieval Validation

Table 4 validates the retrieval subsystem independently.

Table 4. Supporting evaluation: Progressive retrieval component stacking ($N = 100$).

Config.	Hit@1	Hit@3	P@5	R@5	MRR@10
BM25 Lexical	0.530	0.770	0.206	0.617	0.669
Dense Semantic	0.460	0.650	0.176	0.524	0.560
Hybrid RRF	0.520	0.790	0.224	0.662	0.673
Hybrid + Reranker	0.700	0.910	0.254	0.751	0.805
Full Retrieval	0.740	0.920	0.302	0.861	0.833

BM25 records 0.530 Hit@1 and 0.669 MRR@10, compared with 0.460 Hit@1 and 0.560 MRR@10 for dense semantic search. Hybrid RRF achieves 0.790 Hit@3 and 0.520 Hit@1, balancing lexical precision and semantic recall. Adding the cross-encoder reranker (bge-reranker-v2-m3) yields the most pronounced individual retrieval gain, raising Hit@1 from 0.520 to 0.700 (+18.0 pp) and MRR@10 to 0.805. The full retrieval configuration reaches 0.740 Hit@1, 0.861 Recall@5, and 0.833 MRR@10. Because this final configuration incorporates domain-specific evidence components, the difference cannot be attributed to one component alone. In live deployment, the retrieval subsystem is complemented by Redis-backed session caching to maintain dialogue history and

rapidly supply conversational context to the supervisor, minimizing database round-trips while isolating benchmark retrieval evaluation to single-turn precision.

5.5 Failure Analysis

Table 5 categorizes error modes across the 58 failure instances ($< 50\%$ Fact Coverage) in Scenario 1 (34 for Single Agent, 24 for CTU-Chat), established through manual qualitative review of individual execution traces and intermediate outputs.

Table 5. Failure taxonomy across architecture configurations ($N = 100$). Categories are non-mutually exclusive; a single query may exhibit multiple error modes.

Category	Failure Mode	S1-A	S1-C	Category	Failure Mode	S1-A	S1-C
Routing	Corrected misroute	N/A	1	Tool	Invalid arguments	4	1
	Unrecoverable misroute	N/A	3		Wrong-family call	3	1
Retrieval	Gold source absent	3	3	Generation	Unnecessary call	5	1
	Cross-domain omission	3	2		Result ignored	2	1
	Reranker demoted	7	6		Partial fact omission	22	14
Total Failed Queries ($< 50\%$ Fact):				S1-A (Single): 34		S1-C (CTU): 24	

CTU-Chat reduces overall failure instances from 34 to 24. Tool errors drop from 12 in Single Agent (4 invalid arguments, 3 wrong-family calls, 5 unnecessary invocations) to 3 in CTU-Chat due to scoped registries. Retrieval omissions (3 gold absent, 7 vs. 6 partial demotions) affect both configurations similarly because pre-generation retrieval was matched. In generation, partial fact omissions under complete retrieval ($sr = 1.0$) decrease from 22 in Single Agent to 14 in CTU-Chat, reflecting more structured synthesis in specialist prompt contracts.

Trace-level root cause analysis of the three unrecoverable routing failures reveals two primary failure modes. First, semantic ambiguity occurs at domain boundaries. For example, administrative queries regarding major-transfer procedures contained explicit curriculum keywords, causing the supervisor to misroute them to the academic specialist whose scoped registry lacked regulatory handbooks. Second, multi-intent structural collapse arises under single-owner dispatch. Specifically, hybrid queries requesting both curriculum requirements and tuition schedules were assigned to a single specialist without a collaborative handoff mechanism, omitting the secondary domain.

6 Discussion

6.1 Architectural Findings and Operational Trade-offs

RQ1 (Architecture-Level Effectiveness): The architecture-level comparison indicates that supervisor-routed multi-agent decomposition achieves a 10.0 pp higher Automated Fact Pass Rate ($\text{Pass}_{\geq 50\%}$) point estimate than the Unified Single Agent (paired bootstrap

CI [+2.0, +19.0] pp; exact McNemar $p=0.0414$). This gain is accompanied by higher Fact Coverage (+3.58 pp, 95% CI [-1.2, +8.5] pp) and 41.4% fewer recorded specialist input tokens (3,690 vs. 6,296; supervisor adds ~650 routing tokens/query). However, end-to-end dispatch latency exceeds post-retrieval single-agent generation (10,652 vs. 5,159 ms). Routed Generic records lower Fact Coverage (54.2% vs. 61.0%; $\Delta=-6.85$ pp, $p<0.05$), demonstrating that specialist policies contribute substantially to answer fidelity under shared routing.

RQ2 & RQ3 (Mechanisms, Tool Isolation, and Registry Scaling): Component ablations reveal two distinct roles. *Accuracy drivers:* Heterogeneous evidence access (S2-A3) improves Source Recall by +12.3 pp overall (+17.8 pp on finance); specialist policies (S2-A1) govern coverage (-6.85 pp Fact Coverage); route repair (S2-A5) provides cross-domain robustness (-15.0 pp); and calculators (S2-A4) achieve 8/9 numeric exact matches on tuition queries, versus 3/9 without calculators. *Scalability and efficiency:* At 11 tools, removing tool isolation (S2-A2) does not alter pass rate ($\Delta\text{Pass}_{.50} = 0.0$ pp, CI [-6.0, +6.0] pp) but increases input tokens by +1,270 (+34.4%), showing that tool isolation acts as a cost-containment safeguard rather than an accuracy driver at small scales. Under 51-tool registry expansion (Scenario 3), the Routed Specialist sustains 88.0% tool E2E Success with 9.8 visible tools, compared with 78.7% for Global Top-10 (+9.3 pp, CI [+2.0, +17.3] pp) and 83.3% for Full-Registry Single Agent (+4.7 pp, CI [-1.3, +11.3] pp) while using 52.4% fewer input tokens (1,770 vs. 3,722). In tested registries, specialization kept per-agent tool exposure bounded.

Operational Trade-offs and Deployment Guidance: Operational trade-offs reflect distinct boundaries: (1) *Measurement boundaries:* Single-agent timing measures inference post-retrieval (5,159 ms), whereas multi-agent variants measure end-to-end dispatch inclusive of routing (8,921–10,652 ms). (2) *Specialist overhead:* The observed +1,731 ms delta between generic and specialist multi-agent execution coincides with specialist prompt instructions producing +48% additional output tokens (2,092 vs. 1,415) for domain-tailored reasoning, alongside an estimated upstream supervisor routing overhead of ~650 input tokens. (3) *Deployment guidance and cross-campus transfer:* Monolithic single-agent designs suit low-latency, single-domain queries with small tool pools (≤ 10). In contrast, supervisor-routed multi-agent architectures are justified when counseling demands heterogeneous evidence representations, deterministic arithmetic fidelity, and tool isolation under expanding registries. In principle, institutional counseling domains across higher-education institutions exhibit comparable data structures: relational degree requirements (amenable to property graphs), fee schedules (amenable to deterministic computation), and narrative policies (amenable to vector retrieval). Adapting CTU-CHAT to other university environments is designed to preserve the underlying orchestration graph and contract-dispatch mechanisms without architectural restructuring, primarily requiring the population of corresponding local stores (Neo4j, PostgreSQL, Qdrant) and prompt calibration for domain specialists.

6.2 Relation to Prior Work

While ReAct [19] and Toolformer [15] integrate tools within single agents, full-registry single agents suffer from prompt expansion as the tool registry scales. Unlike collaborative frameworks (e.g., AutoGen [17]) where conversational chatter compounds la-

tency, CTU-CHAT enforces bounded single-owner dispatch with route repair. Ablation S2-A3 highlights text retrieval limits on relational curricula, corroborating GraphRAG’s rationale for heterogeneous evidence [2]. Unlike Adaptive-RAG’s linguistic complexity branching [5], CTU-CHAT routes queries by institutional data modality.

6.3 Threats to Validity

Internal and Operational: Sequential pipeline execution results in an average end-to-end latency of 10.65 s (compared with 5.16 s for post-retrieval generation), presenting an operational limitation for interactive real-time counseling. This overhead stems from serial LLM invocations across routing, contract verification, neural reranking, and generation. Serving variations and network transport introduce additional operational confounds. **External:** While the tripartite data ontology remains invariant across higher education, benchmark queries currently reflect Can Tho University statutes. Cross-campus validation across bilingual or trimester curricula remains future work. Semantic neighbors in Scenario 3 are controlled synthetic constructs. **Construct:** Automated substring matching ($\text{Pass}_{\geq 50\%}$) evaluates lexical phrase occurrence rather than computational validity (e.g., in query HOUT-DIR-FIN-03, regulatory keywords matched 50% coverage despite an incorrect tuition calculation of 192.183M vs. 40M VND); this is mitigated via deterministic tool validation (Scenario 3) and bursar exact-match audits ($N=9$). **Statistical:** Paired bootstrap resampling uses 10,000 iterations ($N=100$); arithmetic exact match ($N=9$) has a small sub-cohort, though the 8/9 vs. 3/9 delta is substantial.

7 Conclusion and Future Work

This study evaluated CTU-CHAT, a supervisor-routed multi-agent RAG framework for university counseling, across architecture comparison, ablation, and tool ambiguity stress testing. Three principal findings address the research questions: (1) CTU-CHAT achieves a 10.0 pp higher Automated Fact Pass Rate ($\text{Pass}_{\geq 50\%}$) point estimate than the matched single-agent baseline (76.0% vs. 66.0%; paired CI [+2.0, +19.0] pp; exact McNemar $p=0.0414$) with 41.4% fewer specialist input tokens; (2) removing specialist policies reduces pass rate by −10.0 pp, while replacing heterogeneous evidence reduces Source Recall by −12.3 pp; and (3) the Routed Specialist sustains 88.0% tool E2E Success at 51 tools (vs. 83.3% and 78.7% for baselines). Domain policies, representation-specific evidence paths, and deterministic calculation appear to contribute to counseling reliability, while scoped tool access provides scalability safeguards under registry expansion. Future work focuses on addressing the observed 10.65 s coordination latency through speculative parallel dispatch, lightweight classifier routing, and semantic caching, alongside evaluating cross-campus transferability across institutions with differing academic calendars.

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